GENERATIVE AI SYSTEMS · UG3 / UG4 · MODULES 1–4

Group Project Guidelines & Judging

How every group project is built, submitted, presented and marked. Your project brief tells you *what* to build; this document tells you *how* it will be assessed.

Team 3–4 students | **Deliverable** 1 Jupyter notebook | **Viva** 10-min presentation + 5-min Q&A | **Marks** 100

SECTION 01

Common Rules for Every Project

Each brief specifies which techniques your project needs: for example retrieval (RAG), a single agent or a multi-agent system, tools, fine-tuning, or advanced prompting. The rules below apply to whichever of these your brief requires.

Area	Rule
Data	Use the public Hugging Face dataset named in your brief, downloaded inside the notebook with `load_dataset`. Subsets are fine if chosen with a fixed seed and their size is stated. Never paste data into the notebook by hand.
Model	Use the open model named in your brief (small Qwen models through Hugging Face or Ollama). A similar open model is allowed with a written justification. No paid API is required. Ollama also runs on Colab: install it with `curl -fsSL https://ollama.com/install.sh | sh`, start `ollama serve` in the background, then `ollama pull <model>`.
Required components	Implement every component your brief marks as required, and evaluate each one on its own as well as within the whole system. Optional components count as stretch goals.
Design choices	Justify your main design decisions (for example single vs multi-agent, retrieval settings, prompting strategy, fine-tuning setup) by comparing them with at least one simpler alternative.
Evaluation	A fixed test subset, at least one baseline and one ablation, a single results table, and an error analysis of at least 10 failures.

Compute	The full notebook must run top to bottom ("Restart & Run All") in under 60 minutes on a free Colab T4 or a laptop, loading saved artefacts (trained adapters, indexes) where training would take longer. Provide a `QUICK_RUN = True` flag that evaluates on a small sample.
Optional app	A Streamlit or Gradio front end is welcome but never required. The notebook is the graded deliverable and must contain the full system, the evaluation and the demo cell even if you also build an app. An app may be used for the live demo; it earns no extra marks on its own.
Integrity	Cite every dataset, model, paper, course lab and tool you use. Mark AI-assisted code. Report results honestly, including negative ones. Do not share code or results with other teams.

SECTION 02

The Notebook: One File, Fixed Structure

Everything is submitted as **one Jupyter notebook**, starting from the provided `Project_Notebook_Template.ipynb`. Keep these sections in this order:

#	Section	Must contain
0	Title & team	Project code, title, members, and a contribution table (who did what)
1	Problem statement	The problem in your own words, the research question, and what success looks like
2	Setup	`%pip install` with versions, imports, seeds, a config cell (models, parameters, paths, `QUICK_RUN`)
3	Data	`load_dataset`, exploration (sizes, examples, label balance), the fixed train/test subsets
4	Baseline	The simplest system named in your brief, evaluated on the test subset
5	System	One subsection per component your brief requires (for example 5a retrieval, 5b agents and tools with the graph drawn, 5c fine-tuning)
6	Evaluation	One results table (baseline, full system, ablations) and at least one chart
7	Error analysis	At least 10 failure cases shown and categorised, with what you would change

8	Demo	A cell with an `ask(...)` function used live in the presentation (an optional app may call the same function)
9	Limitations & ethics	Bias, privacy, misuse and failure risks of your system
10	References	Datasets, models, papers, labs and tools used

> **Submit the notebook with all outputs saved.** Judges read the outputs first; a notebook without outputs, or one that fails on "Restart & Run All", loses the reproducibility marks.

SECTION 03

Presentation: 10 Minutes + 5 Minutes Q&A

Time	Content
1 min	The problem and why it matters
2 min	Data and system design
3 min	Live demo from the notebook's demo cell: one success and one failure
3 min	Results table, ablation and the most interesting finding
1 min	Limitations and what you would do next
5 min	Q&A: every team member must answer at least one question

Slides are optional: you may present directly from the notebook. A hard stop is enforced at 10 minutes.

SECTION 04

Judging Rubric (100 marks)

A. Notebook — 60 marks

Criterion	Marks	Excellent (full marks)	Weak (low marks)

Problem & data	8	Clear problem and research question; data loaded from Hugging Face, explored and split correctly	Vague problem; data not explored or not loaded in the notebook
Implementation of required components	22	Every component the brief requires (e.g. retrieval, agents and tools, fine-tuning, structured prompting) is built correctly, measured on its own, and integrated into a working system	Components missing, superficial, or not measured
Design choices & depth	8	Main choices justified against simpler alternatives; evidence of going beyond the lab code	Lab code copied with minimal adaptation; no justification
Evaluation	12	Fixed test set, baseline, full system and ablation in one table; appropriate metrics; honest reading of results	Only a few hand-picked examples; no baseline
Error analysis & ethics	5	At least 10 failures categorised with causes; concrete limitations and risks	No failure cases; generic ethics paragraph
Reproducibility & clarity	5	Runs top to bottom; seeds, config cell, QUICK_RUN flag; clear markdown narrative	Fails to run; unexplained code cells

B. Presentation — 25 marks

Criterion	Marks	What judges look for
Story & clarity	8	Problem → design → results → insight, within 10 minutes
Live demo	9	Works from the notebook; shows how the system reaches its answer; includes an honest failure
Results & insight	8	Numbers explained, not just shown; a clear takeaway from the ablation

C. Q&A — 15 marks

Criterion	Marks	What judges look for
Technical understanding	8	Correct, specific answers about the team's own design choices and code
Individual contribution	7	Every member answers at least one question about their own part, consistent with the contribution table

Adjustments

- **Individual marks:** the team mark may be adjusted by up to ±10% per student based on the contribution table and Q&A.
- **Penalties:** dataset not from Hugging Face or not loaded in the notebook (−10); notebook submitted without outputs or fails to run (−5 to −10); late submission (−10% per day).
- **Bonus (up to +5, capped at 100):** a completed stretch goal from the brief.

SECTION 05

Suggested Timeline

Week	Milestone
1	Teams formed; project allocated
2	Data loaded and explored; baseline running (notebook sections 0–4)
4	First required component built and measured
6	All required components integrated; full evaluation (sections 5–6)
7	Error analysis, demo cell, limitations; notebook frozen (sections 7–10)
8	10-minute presentation + 5-minute Q&A